

```
# =====
# TEXT SUMMARIZATION TOOL
# Works in Google Colab
# =====

# Install required libraries
!pip install sumy nltk -q

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('punkt_tab')

from google.colab import files
from IPython.display import display
from ipywidgets import widgets

from sumy.parsers.plaintext import PlaintextParser
from sumy.nlp.tokenizers import Tokenizer
from sumy.summarizers.lsa import LsaSummarizer

# =====
# FUNCTION FOR SUMMARIZATION
# =====

def summarize_text(text, sentence_count=5):

    parser = PlaintextParser.from_string(text, Tokenizer("
    summarizer = LsaSummarizer()

    summary = summarizer(parser.document, sentence_count)

    summarized_text = ""

    for sentence in summary:
        summarized_text += str(sentence) + " "

    return summarized_text

# =====
# OPTION 1 : MANUAL TEXT INPUT
# =====
```

```

print("\n=====")
print("TEXT SUMMARIZATION TOOL")
print("=====\n")

choice = input("Choose Input Method:\n1. Upload TXT File\n")

# =====
# FILE UPLOAD OPTION
# =====

if choice == '1':

    print("\nUpload a .txt file")

    uploaded = files.upload()

    file_name = list(uploaded.keys())[0]

    with open(file_name, 'r', encoding='utf-8') as file:
        text = file.read()

# =====
# MANUAL INPUT OPTION
# =====

elif choice == '2':

    print("\nPaste your text below:\n")

    text = input()

else:
    print("Invalid Choice")
    exit()

# =====
# SUMMARY LENGTH INPUT
# =====

print("\nOriginal Text Length:", len(text.split()), "words")

num_sentences = int(input("\nEnter number of sentences for"))

# =====

```

```

# GENERATE SUMMARY
# =====

summary = summarize_text(text, num_sentences)

# =====
# OUTPUT
# =====

print("\n=====")
print("SUMMARY")
print("=====\\n")

print(summary)

print("\n=====")
print("SUMMARY GENERATED SUCCESSFULLY")
print("=====")

# =====
# SAVE SUMMARY TO FILE
# =====

save_option = input("\nDo you want to save summary to txt

if save_option.lower() == 'y':

    with open("summary.txt", "w", encoding='utf-8') as f:
        f.write(summary)

    files.download("summary.txt")

    print("\nsummary.txt downloaded successfully!")

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.

=====
TEXT SUMMARIZATION TOOL
=====

Choose Input Method:
1. Upload TXT File
2. Enter Text Manually

```

Enter 1 or 2: 2

Paste your text below:

AI systems are trained on massive datasets. They analyze this data to identify

Original Text Length: 164 words

Enter number of sentences for summary: 10

=====

SUMMARY

=====

AI systems are trained on massive datasets. They analyze this data to identify

=====

SUMMARY GENERATED SUCCESSFULLY

=====

Do you want to save summary to txt file? (y/n): n

Start coding or generate with AI.

Start coding or generate with AI.